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MARCH 29, 1954
SUPERSEDING
Int. Fed. Spec. GGG-C-00749 (Navy-Ships)
4 June 1952

FEDERAL SPECIFICATION

CUTTER, CABLE, HAND OPERATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers cable cutters used primarily for cutting armored and lead sheathed cables (except submarine cable) that are permanently clamped to surfaces.

1.1.1 *Federal specification coverage.*—This specification does not cover all types of cutters, but only those generally used by the Federal Government.

1.2 Classification.—Cable cutters shall be of the following types as specified (see 6.1):

Type I.—Ratchet and gear.
Type II.—Hydraulic.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-B-591—Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621—Boxes, Wood, Nailed and Lock-Corner.
NN-B-631—Boxes, Wood, Wirebound, for Domestic Shipment.
LLL-B-631—Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-601—Boxes, Wood, Cleated-Plywood.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Non-Government activities may obtain copies of Federal Specifications and Standards upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Copies of the Index of Federal Specifications and Standards may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices.

(Single copies of this specification and other product specifications required by non-Government activities for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Government activties may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards through departmental channels from the established sources of supply.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The quality of the materials, methods of manufacture, and the treatment of parts shall be such as to produce cable cutters conforming to this specification. The material shall be free from defects and shall be of the lightest weight consistent with the strength required for application.

3.1.1 *Fasteners.*—All machine, cap and set screws, nuts, bolts, studs, tap rivets, or other fasteners shall be of corrosion resisting material.

3.2 General design.—The cable cutter shall be of safe, sturdy construction and shall have strong and rugged working parts. Cable cutters shall consist of a cutter head, and a ram and anvil, with a suitable means to maintain pres-

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sure on the ram and anvil when cutting cable. The cutter head shall be C or U shaped, containing a renewable steel blade.

3.3 **Weight.**—The weight of the complete cable cutter shall not exceed 29 pounds.

3.4 **Cutting blade.**—The cutting blade shall be of good-quality tool steel, properly hardened and tempered, showing a hardness of not less than 58 nor more than 62 on a Rockwell C scale. The cutting edge shall be suitably ground for cutting through steel, lead, aluminum, or bronze armor, wax-impregnated fibrous material, and stranded copper conductor. The cutting-edge angle shall be a 30° included angle with a base three-fourths of an inch wide. The cutting blade shall fit neatly into the clamp, in such manner that it may be easily removed for resharpening and replaced. One extra cutting blade shall be supplied with each cable cutter.

3.5 **C or U clamp.**—A C or U clamp shall be formed by one of the parts which surround the cable for cutting. Suitable means shall be provided in the clamp to accommodate and hold the cutting blade in position during the cutting operation.

3.6 **Cable-cutting capacity.**—The cable cutter shall be capable of cutting the largest armored cable which is approximately 2½ inches in diameter.

3.7 **Cutting speed.**—The speed of operation of the tool shall be such that time required to cut the largest cable and to reset the tool for another cut shall be not greater than 120 seconds.

3.8 **Grounding clamp.**—A grounding clamp shall be provided on the cutter which will be suitable for connector copper size number 6AWG.

3.9 **Finish.**—Each cable cutter shall be smoothly finished with the exterior surfaces painted with at least one coat of gray paint satisfactory to the activity concerned.

3.10 **Repair parts.**—The successful bidders supplying cable cutters covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards and clearances in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.11 **Nameplate.**—A nameplate shall be permanently attached to each cable cutter and shall include the manufacturer's name or trademark, type or model number, in order that the source of manufacture may be readily determined.

3.12 **Type I, ratchet and gear.**

3.12.1 *Operation and functional design.*—The cable cutter shall be known as ratchet and gear, and the operation shall be accomplished through hand drive of continuous rotating movement. The cable cutter shall be composed of the following essential parts: a ratchet wheel and pawl, worm and worm wheel, and screw with suitable engaging threads in the worm wheel. The torque required for cutting a 2½-inch-diameter armored cable shall be not greater than 30 foot-pounds. To obtain the adequate torque, the hand drive may be adjustable, but the length of the arm when fully extended shall not exceed 10 inches.

3.12.2 *Dimensions.*—Dimensions A, B, and C as shown on figure 1 shall not exceed 9, 5, and 27 inches, respectively.

3.12.3 *Ratchet mechanism.*—The ratchet gear shall be of steel. The pawl or detent shall be made of steel properly heat treated and shall engage the teeth of the ratchet wheel when under load. Means shall be provided to change the direction of rotation.

3.12.4 *Worm and worm wheel.*—The worm and worm wheel shall be machined from steel or bronze.

3.12.5 *Lubrication.*—The worm mechanism shall operate in oil or grease. A grease fitting or access plate shall be provided in the worm housing for lubricating the worm mechanism.

3.12.6 *Bearings and bushings.*—Thrust bearings shall be of ball type. Bushings shall be made of bronze or other suitable material.

3.12.7 *Screw.*—The screw shall be made of steel; the threads shall be carefully cut and finished. The threaded portion of the screw shall engage with the threads in the worm wheel but shall rotate freely through the bearings and bushings.

3.12.8 Type I cable cutter shall be similar to figure 1.

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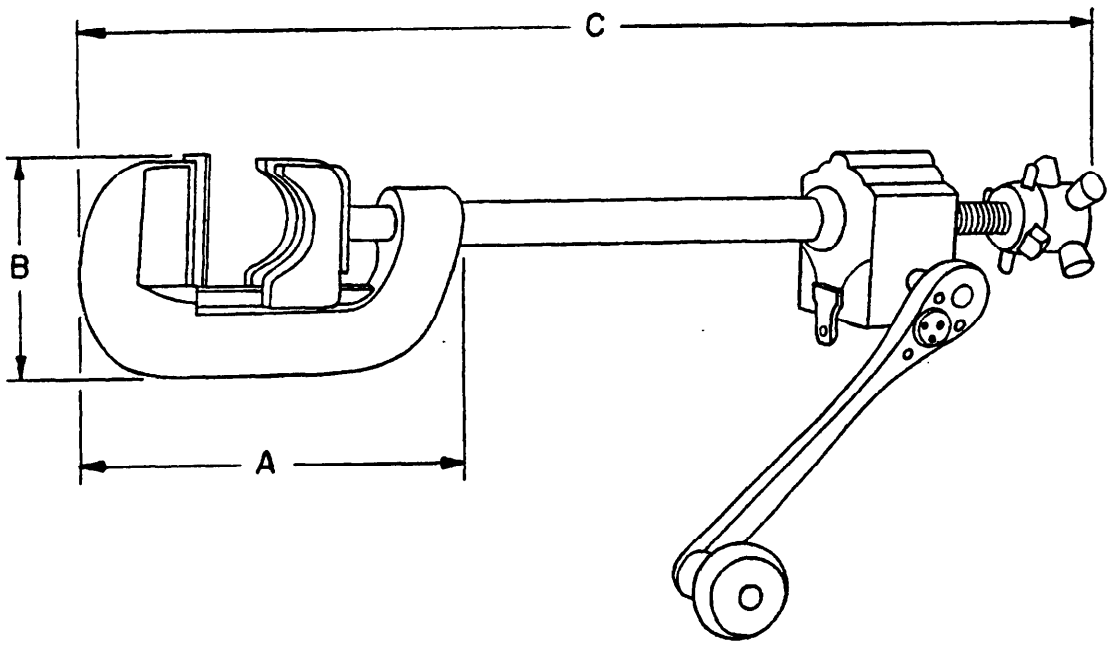


FIGURE 1.—Type I, ratchet and gear.

3.13 Type II, hydraulic.

3.13.1 *Operation and functional design.*—The cable cutter shall be known as hydraulic and the operation shall be accomplished through continuous hand pumping. The cutter shall be composed essentially of a pump and the necessary valves and allied mechanism to provide a compact hydraulic unit which will operate easily and speedily and develop a minimum pressure of 9,000 p. s. i. A valve release shall be provided for the return of the anvil upon completion of the cut.

3.13.2 *Dimensions.*—Dimensions A, B, and C as shown on figure 2 shall not exceed 8 5/16, 7 1/2, and 19 inches, respectively.

3.13.3 *Cylinder and piston.*—The cylinder and piston shall be compactly constructed and shall operate efficiently. All working parts of the hydraulic system shall be entirely enclosed and suitably protected against wear and damage. Suitable means shall be provided when operating in any position to prevent leakage of any kind from the hydraulic system under maximum working pressures, or when idle. The piston and cylinder shall be made from high-grade steel and shall be accurately machined to insure a proper fit between the piston and bore of the cylinder.

3.13.4 *Hydraulic fluid.*—Hydraulic fluid shall not gum or clog the valves under any operating conditions, shall not corrode the steel ram or cylinder, or be injurious to the packings. The hydraulic fluid shall operate efficiently at 100° F. and minus 25° F. or lower with the system filled to normal operating level.

3.13.5 Type II cable cutter shall be similar to figure 2.

3.14 Workmanship.—Workmanship shall be of the highest grade throughout and equal in every respect to the best commercial practice. The cable cutter shall be free from rust, burrs, and any other imperfection which may impair serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All cable cutters of the same type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for inspection.*

4.1.2.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of cutters shall be selected in accordance with table 1 from each inspection lot of material offered

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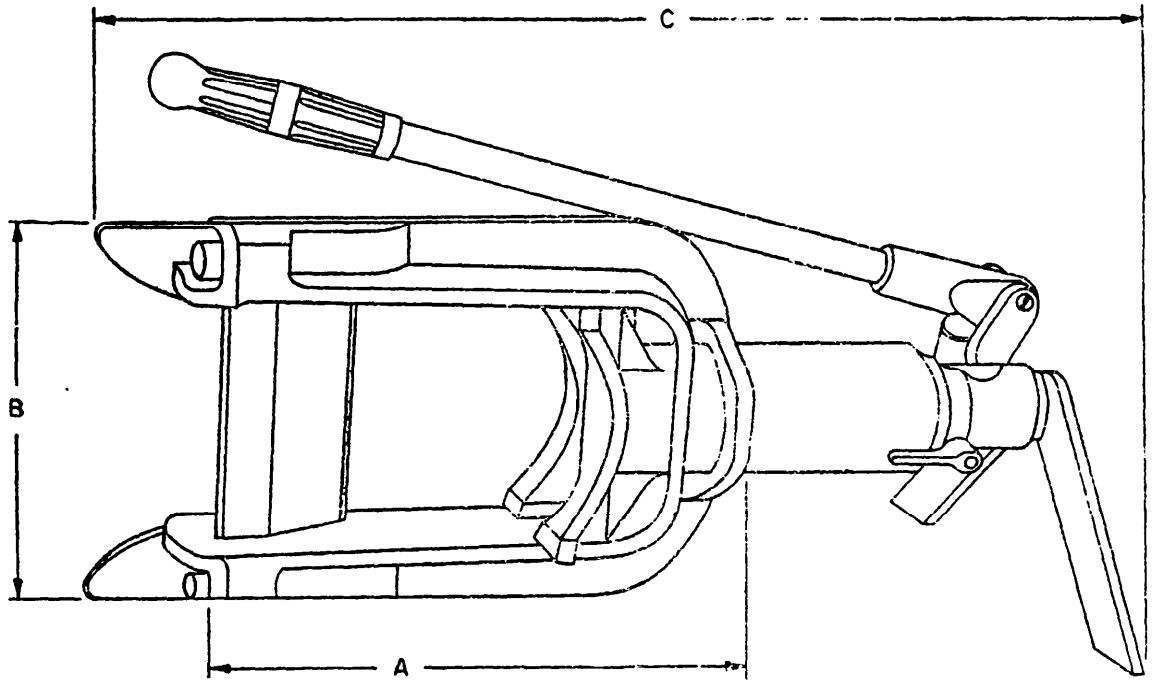


FIGURE 2.—Type II, hydraulic cable cutter.

for Government inspection of visual, dimensional and weight characteristics with lot acceptance based on the single sampling inspection requirements in accordance with table I and the procedures of Military Standard MIL-STD-105.

TABLE I.—*Sampling for visual, dimensional and weight inspection AQL (approx.)=1.5 percent defective*

Number of cutters in an inspection lot	Number of cutters in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL-STD-105.

4.1.2.2 *Sampling for continuous production.*—Where there is continuous production for the Government and inspection lots include at least 200 items, sampling for Government inspection shall be conducted in accordance with Military Standard MIL-STD-105 at inspection level I and $AQL=1.5$ percent defective; the type of sampling (single, double, or multiple) to be determined by the Government inspector.

4.1.3 *Sampling for lot acceptance tests (at a Government laboratory).*—From each lot which has passed inspection at the place of manufacture, the Government inspector shall select the number of samples shown in table II and forward these samples for the tests specified in 4.4 to a Government laboratory designated by the activity concerned.

TABLE II.—*Sampling for lot acceptance tests*

Number of cutters in an inspection lot	Number of cutters in the sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	3	0	1
41 to 110	5	0	1
111 to 300	7	0	1
301 and over	10	0	1

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4.2 **Inspection.**

4.2.1 *Visual and dimensional inspection.*—Each of the sample cutters selected in accordance with table I shall be visually and dimensionally inspected and weighed by the Government inspector to verify compliance with this specification. Any cutter in the sample containing one or more visual or dimensional defects or exceeds the weight requirement shall be rejected, and if the number of defective cutters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Lot acceptance tests (at a Government laboratory).**—Each of the sample cutters selected in accordance with table II shall be subjected to each of the tests specified in 4.4. Any cutter in the sample failing to conform to the test requirements shall cause rejection of the lot represented by the sample.

4.4 **Test procedure.**

4.4.1 *Hardness test.*—The cutting blade of each sample cutter selected in accordance with 4.1.3 shall be subjected to a hardness test to determine conformance with 3.4.

4.4.2 *Performance test.*—Each sample cutter selected in accordance with 4.1.3 shall be tested to determine conformance with 3.6, 3.7, 3.12.1, 3.13.1, and 3.13.3.

4.5 **Rejected lots.**—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming items. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

4.6 **Additional inspection.**—Where other specifications form a part of this specification, sampling, inspection, and tests shall be conducted as required in the referenced specification.

5. PREPARATION FOR DELIVERY

5.1 **Preservation.**—Unless otherwise specified, cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.2 **Packaging.**—Unless otherwise specified, commercial packages will be acceptable.

5.3 **Packing.**—Unless otherwise specified, cutters shall be packed in snug-fitting wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated fiber or solid fiber boxes conforming to Federal Specification NN-B-591, PPP-B-601 (domestic type), NN-B-621, NN-B-631, LLL-B-631, or LLL-B-636, respectively. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.4 **Marking.**

5.4.1 *Interior containers.*—Unless otherwise specified, packages and interior containers shall be marked with the name of the material and contractor's name or trade-mark of such known character that the source of supply may be readily determined.

5.4.2 *Exterior containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein and procurement documents should specify the following (see 5.1, 5.2, 5.3, 5.4.1, and 5.4.2):

(*a*) Title, number, and date of this specification.

(*b*) Type of cable cutter required (see 1.2).

(*c*) Whether preparation for domestic or overseas shipment is required (see sec. 5 and 7.2.3).

6.2 THE ILLUSTRATIONS HEREIN ARE DECRIPTIVE AND NOT RESTRICTIVE, and are not intended to preclude the purchase of cable cutters otherwise conforming to this specification.

6.3 **Transportation description.**—Transportation description applicable to these items is:

Tools, not otherwise indexed by name. Carload weight 30,000 pounds.

Mechanics' hand tools, not otherwise indexed. Truckload volume weight 30,000 pounds.

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Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard of the issue in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specification and standard.*

Military Specification:

MIL-P-15424—Packaging of Hand Tools; General Specification for Domestic and Overseas Preservation.

Military Standard:

MIL-STD-130—Identification Marking of U. S. Property.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Marking for Identification.*—Cable cutters shall be marked in accordance with Military Standard MIL-STD-130.

7.2.3 *Preparation for delivery for domestic or overseas shipment.*—Cutters shall be cleaned, preserved, unit protected, packaged, packed, and marked in accordance with Military Specification MIL-P-15424. The nomenclature shall be as follows:

Cutter, Cable, Hand-operated.

MILITARY INTEREST:
Navy—Sh S Y.
Army—E Sig T.
Air Force.

Copies of this specification may be purchased for 5 cents.

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